Production enhances prediction in comprehension: Evidence from ERPs

Miriam Schulz[1], Masato Nakamura[1], Francesca Delogu[1], Matthew W. Crocker[1]
[1]Saarland University
mschulz@lst.uni-saarland.de

Comprehenders routinely generate predictions during language comprehension, but the underlying mechanisms remain unclear. The prediction-by-production hypothesis suggests that comprehension relies on the production system [5, 6], with studies showing facilitation of highly predicted words by modulating the involvement of production processes [1, 2, 3]. It is still uncertain, however, whether increased engagement of the production system benefits all predictable words, i.e. both high and low cloze continuations, or only the most expected ones in a "winner-takes-all" fashion.

We report ERP results (N=36 native German participants) from an experiment investigating the strength of expectancy effects contrasting two tasks: standard "Reading-for-Comprehension" (RfC), where each trial is followed by a comprehension question, and a novel "Reading-for-Production" (RfP) paradigm, where each trial ends with a spoken sentence completion (cloze-style) task after at least two post-critical regions, encouraging readers to prepare for production during comprehension (Figure 1). Trials began with a context sentence, followed by a target sentence containing a noun manipulated for expectancy: *HC* (high cloze), *LC* (low cloze but attested), and *IM* (implausible; Table 1). If covert production facilitates prediction across the board, we expect a similar enhancement in facilitation (i.e., a stronger reduction in N400 amplitude) in RfP for both *LC* and *HC* over *IM*. By contrast, if covert production pre-activates only the best target candidates, facilitation in RfP should be greater for *HC vs. LC,* but not for *LC vs. IM*. In the post-N400 time window, we might additionally expect larger positivities for less predicted targets in RfP compared to RfC, reflecting failed prediction/integration costs [8].

A linear mixed-effects (LME) analysis of the N400 time window (300-500ms) reveals main effects of expectancy (larger negativities for both *LC vs. HC* and *IM vs. LC,* both $p<.001$) and task (more positive amplitudes overall in RfP, $p<.001$; see Figure 2A). Crucially, interaction terms reveal stronger facilitation in RfP compared to RfC for both *LC vs. IM* ($p=.006$) and *HC vs. LC* ($p<.001$). In the post-N400 time window (600-1000ms), where RfP again resulted in overall stronger positivities ($p=.014$), *LC* targets elicited a greater positivity than *HC* continuations ($p=.017$), with an interaction term further showing stronger positivities for *IM vs. LC* in RfP ($p<.001$; see Figure 2B).

In line with findings from a previous SPR study using the same paradigm, the ERP results show that engaging production mechanisms strengthens predictions. Our ERP results reveal a strong processing advantage for the most likely target candidate, as indexed by greater attenuation of the N400, together with enhanced late positivities for implausible targets. These findings suggest both that the involvement of production during comprehension can be modulated by the task, and further that increased engagement of the production system disproportionately enhances expectations for the most likely words.

Context sentence (displayed in full): Bobby will einen Zettel am Kühlschrank befestigen.
Bobby wants to attach a note to the refrigerator.

Target sentence (presented word by word):

Condition	Start	Pre-critical	Target	Spill 1	Spill 2	Production/End
HC: High cloze	Er nimmt einen *He takes a*	kleinen *small*	**Magneten** ***magnet***	aus *out of*	der *the*	| Abstellkammer. | *storage room.*
LC: Low cloze	Er nimmt einen *He takes a*	kleinen *small*	**Hocker** ***stool***	aus *out of*	der *the*	| Abstellkammer. | *storage room.*
IM: Implausible	Er nimmt einen *He takes a*	kleinen *small*	**Drucker** ***printer***	aus *out of*	der *the*	| Abstellkammer. | *storage room.*

Table 1: Example item. Materials were partially adapted from [4] and [7]. Target nouns were selected to have a cloze probability ranging from 50-95% for *HC* (mean = 78%), 5-16% for *LC* (mean = 6%), and zero cloze for *IM.* Plausibility ratings were collected from N=45 native German participants in a norming pretest (7-point Likert scale) presenting target sentences up to and including the target noun, mean ratings per condition: *HC* = 6.76, *LC* = 5.14, *IM* = 1.56.

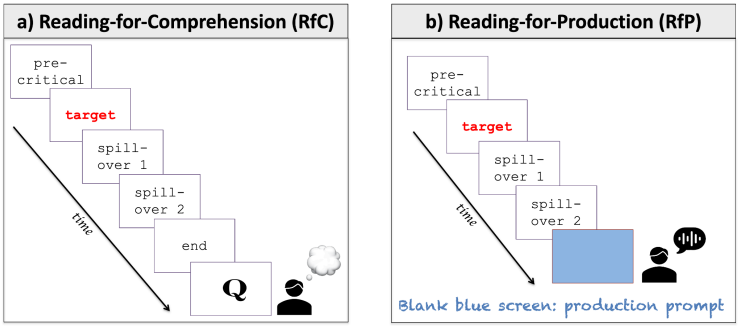

Figure 1: Task regimes: a) *Reading-for-Comprehension (RfC):* a comprehension question following every trial. b) *Reading-for-Production (RfP):* participants read silently until the screen turns blue, and then complete each sentence aloud. Their answers were recorded. All participants completed 90 trials in one and then the other task. Task order was counterbalanced across participants. Short fillers were used to vary the locus of the production prompt in the target sentence.

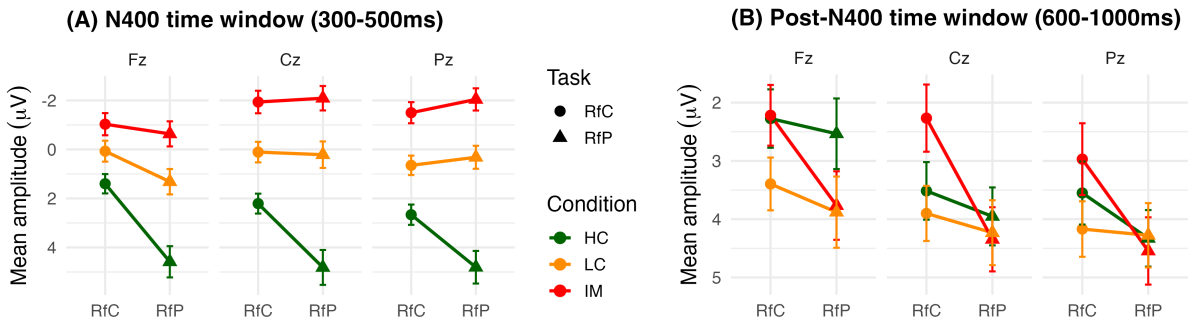

Figure 2: (A): ERP results in the N400 time window (300-500ms). (B): ERP results in the post-N400 time window (600-1000ms). Error bars show ±1SE of the by-subject condition mean.

References

[1] Hintz, Meyer, & Huettig. (2016). *Quarterly Journal of Experimental Psychology.*
[2] Lelonkiewicz, Rabagliati, & Pickering. (2021). *Quarterly Journal of Experimental Psychology.*
[3] Martin, Branzi, & Bar. (2018). *Scientific Reports.*
[4] Nicenboim, Vasishth, & Rösler. (2020). *Neuropsychologia.*
[5] Pickering & Gambi. (2018). *Psychological Bullletin.*
[6] Pickering & Garrod. (2013). *Behavioral and Brain Sciences.*
[7] Stone, Nicenboim, Vasishth, & Rösler. (2023). *Neurobiology of Language.*
[8] Van Petten & Luka. (2012). *International Journal of Psychophysiology.*